MI NIR IQ API

Version 0.3

REVISION HISTORY

Revision No.	Description	Date
0.1	• Initial release	1/20/2022
0.2	• Add IQ API and data type	2/11/2022
0.3	• Add Load IQ bin file API and releated data type	3/7/2022

TABLE OF CONTENTS

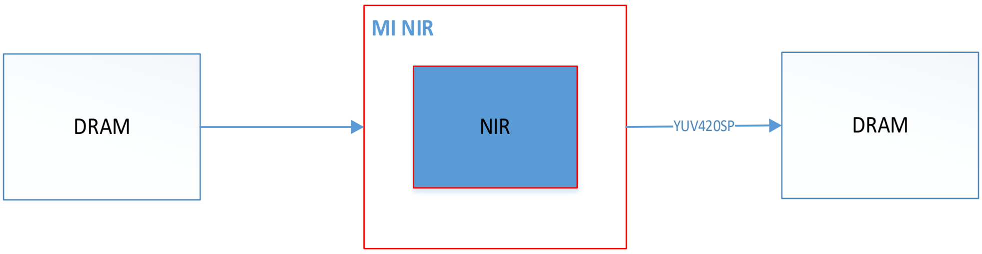
1. OVERVIEW

1.1. Module description

NIR Module is a 2-image fuser to combine a visible image and a near-infrared (NIR) image.

a. It will extract the detail (high contrast) from both images
b. It will reduce the noise level from both images

1.2. Flow chart



For input image format in dram, refer to the description of the following keywords.

1.3. Keyword

Input dram layout:

VIS:
Visible

NIR:
near-infrared

Device:
NIR Hardware device

Channel:
Time division multiplexed channel on device

Input image format:

YUV420SP

Output image format:
YUV420SP

Input resolution:
Muffin:
Maximum support 2688 x 1520
Support input image width 8Pixel alignment, Stride 16 alignment

Output resolution:
Same as input resolution

2. API REFERENCE

This function module provides the following user APIs.

API Name	Function
Type of System	
MI_NIR_IQ_SetBlendingSaturation	Set VIS/NIR Blending Saturation
MI_NIR_IQ_GetBlendingSaturation	Get VIS/NIR Blending Saturation
MI_NIR_IQ_SetContrast	Set NIR Contrast
MI_NIR_IQ_GetContrast	Get NIR Contrast
MI_NIR_IQ_SetSaturation	Set NIR Saturation
MI_NIR_IQ_GetSaturation	Get NIR Saturation
MI_NIR_IQ_SetWeight	Set VIS/NIR Weight
MI_NIR_IQ_GetWeight	Get VIS/NIR Weight
MI_NIR_IQ_ApiCmdLoadBinFile	Load IQ bin

2.1. MI_NIR_IQ_SetBlendingSaturation

- Description
  Set VIS/NIR Blending Saturation.

- Syntax
  MI_S32 MI_NIR_IQ_SetBlendingSaturation (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_BlendingSaturationAttr_t *data);

- Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of blending saturation parameter to be set.	Input

- Return value
  - MI_NIR_OK(0) successful.
  - Non-Zero: failed, see error code for details.

- Dependence
  - Header:mi_nir_iq.h
  - Library:libmi_nir.a / libmi_nir.so

- Related APIs
  MI_NIR_IQ_GetBlendingSaturation

2.2. MI_NIR_IQ_GetBlendingSaturation

- Description
  Get VIS/NIR Blending Saturation.

- Syntax
  MI_S32 MI_NIR_IQ_GetBlendingSaturation (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_BlendingSaturationAttr_t *data);

- Parameter

Parameter name	**Description**	**Input/Output**
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of blending saturation parameter to be get.	Output

- Return value
  - MI_NIR_OK(0) successful.
  - Non-Zero: failed, see error code for details.

- Dependence
  - Header:mi_nir_iq.h
  - Library:libmi_nir.a / libmi_nir.so

- Related APIs
  MI_NIR_IQ_SetBlendingSaturation

2.3. MI_NIR_IQ_SetContrast

- Description
  Set NIR Contrast.

- Syntax

MI_S32 MI_NIR_IQ_SetContrast(MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_ContrastAttr_t *data);

➢ Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of contrast parameter to be set	Input

➢ Return value

- MI_NIR_OK(0) successful.
- Non-Zero: failed, see error code for details.

➢ Dependence

- Header:mi_nir_iq.h
- Library:libmi_nir.a / libmi_nir.so

※ Note

Muffin: Not support

➢ Related APIs

MI_NIR_IQ_GetContrast

2.4. MI_NIR_IQ_GetContrast

➢ Description

Get NIR Contrast.

➢ Syntax

MI_S32 MI_NIR_IQ_GetContrast(MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_ContrastAttr_t *data);

➢ Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of contrast parameter to be get	Output

➢ Return value

- MI_NIR_OK(0) successful.
- Non-Zero: failed, see error code for details.

➢ Dependence

- Header:mi_nir_iq.h
- Library:libmi_nir.a / libmi_nir.so

※ Note
Muffin: Not support

➢ Related APIs
MI_NIR_IQ_SetContrast

2.5. MI_NIR_IQ_SetSaturation

➢ Description
Set NIR Saturation.

➢ Syntax
MI_S32 MI_NIR_IQ_SetSaturation (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_SaturationAttr_t *data);

➢ Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of Saturation parameter to be set	Input

➢ Return value
- MI_NIR_OK(0) successful.
- Non-Zero: failed, see error code for details.

➢ Dependence
- Header:mi_nir_iq.h
- Library:libmi_nir.a / libmi_nir.so

※ Note
Muffin: Not support

➢ Related APIs
MI_NIR_IQ_GetSaturation

2.6. MI_NIR_IQ_GetSaturation

➢ Description
Get NIR Saturation.

➢ Syntax

MI_S32 MI_NIR_IQ_GetSaturation (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_SaturationAttr_t *data);

- Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of saturation parameter to be get	Output

- Return value
  - MI_NIR_OK(0) successful.
  - Non-Zero: failed, see error code for details.

- Dependence
  - Header:mi_nir_iq.h
  - Library:libmi_nir.a / libmi_nir.so

※ Note

Muffin: Not support

- Related APIs

  MI_NIR_IQ_SetSaturation

2.7. MI_NIR_IQ_SetWeight

- Description

  Set VIS/NIR weight.

- Syntax

  MI_S32 MI_NIR_IQ_SetWeight (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_WeightAttr_t *data);

- Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of vis and nir weight parameter to be set	Input

- Return value
  - MI_NIR_OK(0) successful.
  - Non-Zero: failed, see error code for details.

- Dependence

- Header:mi_nir_iq.h
- Library:libmi_nir.a / libmi_nir.so

➢ Related APIs
MI_NIR_IQ_GetWeight

2.8. MI_NIR_IQ_GetWeight

➢ Description
Get VIS/NIR weight.

➢ Syntax
MI_S32 MI_NIR_IQ_GetWeight (MI_U32 DevId, MI_U32 ChnId, MI_NIR_IQ_WeightAttr_t *data);

➢ Parameter

Parameter name	Description	Input/ Output
DevId	Device ID	Input
ChnId	Channel ID	Input
*data	Pointer of vis and nir weight parameter to be get	Output

➢ Return value
- MI_NIR_OK(0)   successful.
- Non-Zero: failed, see error code for details.

➢ Dependence
- Header:mi_nir_iq.h
- Library:libmi_nir.a / libmi_nir.so

➢ Related APIs
MI_NIR_IQ_SetWeight

2.9. MI_NIR_IQ_ApiCmdLoadBinFile

➢ Description
Load IQ bin.

➢ Syntax
MI_S32 MI_NIR_IQ_ApiCmdLoadBinFile(MI_U32 DevId, MI_U32 Channel, char *filepath, MI_U32 user_key);

➢ Parameter

Parameter	Description	Input/

name		Output
DevId	Device ID	Input
Channel	Channel ID	Input
filepath	IQ bin file path	Input
user_key	Customer key	Input

- Return value
  - MI_NIR_OK(0) successful.
  - Non-Zero: failed, see error code for details.

- Return value
  - 头文件：mi_nir_iq.h
  - 库文件：libmi_nir.a / libmi_nir.so

- Related APIs
  N/A

3. DATA TYPE

3.1. Data structure list

Video input related data type definition is as following:

Data Type	Definiton
MI_NIR_IQ_Bool_e	Define IQ Bool Type
MI_NIR_IQ_OpType_e	Define IQ work mode
MI_NIR_IQ_BlendingSaturationAutoAttr_t	Define IQ BlendingSaturaton Auto mode struct.
MI_NIR_IQ_BlendingSaturationManualAttr_t	Define IQ BlendingSaturaton Manual mode struce.
MI_NIR_IQ_BlendingSaturationParam_t	Define IQ BlendingSaturaton struct.
MI_NIR_IQ_BlendingSaturationGain_t	Define IQ BlendingSaturaton gain parameters.
MI_NIR_IQ_BlendingSaturationCurve_t	Define IQ BlendingSaturaton curve parameters.
MI_NIR_IQ_BlendingSaturationAttr_t	Define IQ BlendingSaturaton Attr struct.
MI_NIR_IQ_ContrastAutoAttr_t	Define IQ Contrast Auto mode struct.
MI_NIR_IQ_ContrastManualAttr_t	Define IQ Contrast Manual mode struct.
MI_NIR_IQ_ContrastParam_t	Define IQ Contrast parameters.
MI_NIR_IQ_ContrastAttr_t	Define IQ Contrast Attr struct.
MI_NIR_IQ_SaturationAutoAttr_t	Define IQ Saturation Auto mode struct.
MI_NIR_IQ_SaturationManualAttr_t	Define IQ Saturation Manual mode struct.
MI_NIR_IQ_SaturationParam_t	Define IQ Saturation parameters.

MI_NIR_IQ_SaturationAttr_t	Define IQ Saturation Attr struct.
MI_NIR_IQ_WeightAutoAttr_t	Define IQ VIS and NIR Weight Auto mode struct.
MI_NIR_IQ_WeightManualAttr_t	Define IQ VIS and NIR Weight Manual mode struct.
MI_NIR_IQ_WeightParam_t	Define IQ VIS and NIR Weight parameters.
MI_NIR_IQ_WeightAttr_t	Define IQ VIS and NIR Weight Attr struct.

3.2. MI_NIR_IQ_Bool_e

➢ Description

IQ bool enum。

➢ Definition

```
typedef enum
{
    E_NIR_IQ_FALSE = 0,
    E_NIR_IQ_TRUE  = !E_NIR_IQ_FALSE,
    E_NIR_IQ_BOOL_MAX
} MI_NIR_IQ_Bool_e;
```

➢ Member

Membername	Description
E_NIR_IQ_FALSE	Bool value is 0
E_NIR_IQ_TRUE	Bool value is 1
E_NIR_IQ_BOOL_MAX	Bool Max value

※ Note

N/A

➢ Related data type and interface

N/A

3.3. MI_NIR_IQ_OpType_e

➢ Description

IQ work mode。

➢ Definition

```
typedef enum
{
    E_NIR_IQ_OP_TYP_AUTO   = 0,
    E_NIR_IQ_OP_TYP_MANUAL = !E_NIR_IQ_OP_TYP_AUTO,
    E_NIRNIR_IQ_OP_TYP_MODE_MAX
```

```
} MI_NIR_IQ_OpType_e;
```

➢ Member

Membername	Description
E_NIR_IQ_OP_TYP_AUTO	Auto mode
E_NIR_IQ_OP_TYP_MANUAL	Manual Mode
E_NIR_IQ_OP_TYP_MODE_MAX	Work mode max value

※ Note
N/A

➢ Related data type and interface
N/A

3.4. MI_NIR_IQ_BlendingSaturationGain_t

➢ Description
Define IQ BlendingSaturaton Gain parameters. This is for DSP, not for HW.

➢ Definition

```
typedef struct MI_NIR_IQ_BlendingSaturationGain_s
{
    MI_U8 u8YGain; // 0~255
    MI_U8 u8MaxUVRatio;
} MI_NIR_IQ_BlendingSaturationGain_t;
```

➢ Member

Membername	Description
u8YGain	Luminance gain of vis and nir, u8YGain value smaller, vis ratio higher Value Range:0~255
u8MaxUVRatio	Max UV gain Value. Value Range:0~255

※ Note
Muffin: Support gain adjust。

➢ Related data type and interface
N/A

3.5. MI_NIR_IQ_BlendingSaturationCurve_t

- Description

  Define IQ BlendingSaturaton curve parameters. This is for hw, not for dsp.

- Definition

```
typedef struct MI_NIR_IQ_BlendingSaturationCurve_s
{
    MI_U16 u16CurveX[BLEND_SAT_X_NUM];
    MI_U16 u16CurveY[BLEND_SAT_Y_NUM];
} MI_NIR_IQ_BlendingSaturationCurve_t;
```

- Member

Membername	Description
u16CurveX	Blending Saturation Curve x coordinate ， every value represents the gap between the pre and next value. It's value is Luminance, and it is power of 2. BLEND_SAT_X_NUM = 8. Value Range:0~11
u16CurveY	Blending Saturation Curve Curve y coordinate，the value bigger ,vis ratio highter. BLEND_SAT_Y_NUM = 9. Value Range::0~7:0~4095; 8:-2048~2047

※ Note

Muffin: Not support

- Related data type and interface

  N/A

3.6. MI_NIR_IQ_BlendingSaturationParam_t

- Description

  Define IQ BlendingSaturaton struct.

- Definition

```
typedef struct MI_NIR_IQ_BlendingSaturationParam_s
{
    union
    {
        MI_NIR_IQ_BlendingSaturationGain_t stBlendingSatgain；
        MI_NIR_IQ_BlendingSaturationCurve_t  stBlendingSatLut;
    }
} MI_NIR_IQ_BlendingSaturationParam_t;
```

➢ Member

Membername	Description
stBlendingSatGain	IQ Blending Saturaton gain parameter
stBlendingSatLut	IQ Blending Saturaton curveparameter

※ Note

Muffin: Support gain adjustment. Not support curve adjustment.

➢ Related data type and interface

N/A

3.7. MI_NIR_IQ_BlendingSaturationAutoAttr_t

➢ Description

Define IQ BlendingSaturaton Auto mode struct.

➢ Definition

```
typedef struct BlendingSaturationAutoAttr_s
{
    MI_NIR_IQ_BlendingSaturationParam_t  stParaAPI[MI_NIR_AUTO_NUM];
} BlendingSaturationAutoAttr_t;
```

➢ Member

Membername	Description
stParaAPI[MI_NIR_AUTO_NUM];	Auto Mode parameter struct，MI_NIR_AUTO_NUM = 16 .

3.8. MI_NIR_IQ_BlendingSaturationManualAttr_t

➢ Description

Define IQ BlendingSaturaton manual mode struct.

➢ Definition

```
typedef struct MI_NIR_IQ_BlendingSaturationManualAttr_s
{
    MI_NIR_IQ_BlendingSaturationParam_t stParaAPI;
} MI_NIR_IQ_BlendingSaturationManualAttr_t;
```

➢ Member

Membername	Description
stParaAPI;	Manual Mode struct

※ Note
N/A

➢ Related data type and interface
N/A

3.9. MI_NIR_IQ_BlendingSaturationAttr_t

➢ Description
Define IQ BlendingSaturaton struct.

➢ Definition
```
typedef struct MI_NIR_IQ_BlendingSaturationAttr_s
{
    MI_NIR_IQ_Bool_e                        bEnable;
    MI_NIR_IQ_OpType_e                      enOpType;
    MI_NIR_IQ_BlendingSaturationAutoAttr_t  stAuto;
    MI_NIR_IQ_BlendingSaturationManualAttr_t  stManual;
} MI_NIR_IQ_BlendingSaturationAttr_t;
```

➢ Member

Membername	Description
bEnable	Trigger of Blending saturation settings. OFF:E_NIR_IQ_FALSE = 0 ON:E_NIR_IQ_TRUE = 1
enOpType	Blending saturation work mode setting. Auto Mode:E_NIR_IQ_OP_TYP_AUTO =0 Manual Mode:E_NIR_IQ_OP_TYP_MANUAL =1
stAuto	NIR Blending Saturaton auto mode struct
stManual	NIR Blending Saturaton manual mode struct

※ Note
N/A

➢ Related data type and interface
MI_NIR_IQ_SetBlendingSaturation
MI_NIR_IQ_GetBlendingSaturation

3.10. MI_NIR_IQ_ContrastParam_t

➢ Description
Define IQ Contrast parameters.

➢ Definition

```
typedef struct MI_NIR_IQ_ContrastParam_s
{
    MI_U16 u16ContrastThe;
    MI_U16 u16ContrastMin;
    MI_U16 u16CurveX[CONTRAST_X_NUM];
    MI_U16  u16CurveY[CONTRAST_Y_NUM];
} MI_NIR_IQ_ContrastParam_t;
```

➢ Member

Membername	Description
u16ContrastThe	NIR Contrast weight threshold Value Range: 0~4095
u16ContrastMin	NIR Contrast weight min vaule Value Range: 0~4095
u16CurveX[CONTRAST_X_NUM]	Contrast x coordinate， every value represents the gap between the pre and next value.It is power of 2. CONTRAST_X_NUM = 8. Value Range:0~11
u16CurveY[CONTRAST_Y_NUM]	Contrast y coordinate，the value sm,aller the nir ratio lower. CONTRAST_Y_NUM = 9. Value Range:0 ~ 6142

※ Note

N/A

➢ Related data type and interface

N/A

3.11. MI_NIR_IQ_ContrastAutoAttr_t

➢ Description

Define IQ Contrast Auto mode struct.

➢ Definition

```
typedef struct MI_NIR_IQ_ContrastAutoAttr_s
{
   MI_NIR_IQ_ContrastParam_t stParaAPI[MI_NIR_AUTO_NUM];
} MI_NIR_IQ_ContrastAutoAttr_t;
```

➢ Member

Membername	Description
stParaAPI[MI_NIR_AUTO_NUM];	Auto mode struct，MI_NIR_AUTO_NUM = 16 .

※ Note
N/A

➢ Related data type and interface
N/A

3.12. MI_NIR_IQ_ContrastManualAttr_t

➢ Description
Define IQ Contrast Manual mode struct.。

➢ Definition

```
typedef struct MI_NIR_IQ_ContrastManualAttr_s
{
    MI_NIR_IQ_ContrastParam_t stParaAPI;
} MI_NIR_IQ_ContrastManualAttr_t;
```

➢ Member

Membername	Description
stParaAPI;	Manual mode struct.

※ Note
N/A

➢ Related data type and interface
N/A

3.13. MI_NIR_IQ_ContrastAttr_t

➢ Description
Define IQ Contrast Attr struct.

➢ Definition

```
typedef struct MI_NIR_IQ_ContrastAttr_s
{
    MI_NIR_IQ_Bool_e                bEnable;
    MI_NIR_IQ_OpType_e              enOpType;
    MI_NIR_IQ_ContrastAutoAttr_t    stAuto;
```

```
    MI_NIR_IQ_ContrastManualAttr_t stManual;
} MI_NIR_IQ_ContrastAttr_t;
```

- Member

Membername	Description
bEnable	Trigger of Contrast settings. OFF:E_NIR_IQ_FALSE = 0 ON:E_NIR_IQ_TRUE = 1
enOpType	Contrast work mode setting. Auto mode:E_NIR_IQ_OP_TYP_AUTO =0 Manual Mode:E_NIR_IQ_OP_TYP_MANUAL =1
stAuto	NIR Auto mode struct
stManual	NIR Manaul mode struct

※ Note

N/A

- Related data type and interface

MI_NIR_IQ_SetContrast
MI_NIR_IQ_GetContrast

3.14. MI_NIR_IQ_SaturationParam_t

- Description

Define IQ Saturation parameters..

- Definition

```
typedef struct MI_NIR_IQ_SaturationParam_s
{
    MI_U16 u16SatConf;
    MI_U16 u16CurveX[SAT_X_NUM];
   MI_U16 u16CurveY[SAT_Y_NUM];
} MI_NIR_IQ_SaturationParam_t;
```

- Member

Membername	Description
u16SatConf	Saturation value. Value Range:0 ~ 6142
u16CurveX[SAT_X_NUM]	Saturation x coordinate，every value represents the gap between the pre and next value.It is power of 2. SAT_X_NUM = 8. Value Range:0~11

u16CurveY[SAT_Y_NUM]	Saturation y coordinate. It represents color saturation. SAT_Y_NUM = 9. Value Range:0 ~ 6142;

※ Note

N/A

➢ Related data type and interface

N/A

3.15. MI_NIR_IQ_SaturationAutoAttr_t

➢ Description

Define IQ Saturation Auto mode struct.

➢ Definition

```
typedef struct MI_NIR_IQ_SaturationManualAttr_s
{
    MI_NIR_IQ_SaturationParam_t stParaAPI[MI_NIR_AUTO_NUM];
} MI_NIR_IQ_SaturationManualAttr_t;
```

➢ Member

Membername	Description
stParaAPI[MI_NIR_AUTO_NUM];	Saturaion auto mode struct， MI_NIR_AUTO_NUM = 16

※ Note

N/A

➢ Related data type and interface

N/A

3.16. MI_NIR_IQ_SaturationManualAttr_t

➢ Description

Define IQ Saturation Manual mode struct.

➢ Definition

```
typedef struct MI_NIR_IQ_SaturationManualAttr_s
{
    MI_NIR_IQ_SaturationParam_t stParaAPI;
} MI_NIR_IQ_SaturationManualAttr_t;
```

➢ Member

Membername	Description
stParaAPI;	Saturation manual mode struct.

※ Note
N/A

➢ Related data type and interface
N/A

3.17. MI_NIR_IQ_SaturationAttr_t

➢ Description
Define IQ Saturation Attr struct.

➢ Definition

```
typedef struct MI_NIR_IQ_SaturationAttr_s
{
    MI_NIR_IQ_Bool_e                   bEnable;
    MI_NIR_IQ_OpType_e                 enOpType;
    MI_NIR_IQ_SaturationAutoAttr_t     stAuto;
    MI_NIR_IQ_SaturationManualAttr_t   stManual;
} MI_NIR_IQ_SaturationAttr_t;
```

➢ Member

Membername	Description
bEnable	Trigger of Saturation settings. OFF:E_NIR_IQ_FALSE = 0 ON:E_NIR_IQ_TRUE = 1
enOpType	Saturation work mode setting. Auto mode:E_NIR_IQ_OP_TYP_AUTO = 0 Manual mode:E_NIR_IQ_OP_TYP_MANUAL = 1
stAuto	NIR Saturation Auto mode struct
stManual	NIR Saturation Manual mode struct

※ Note
N/A

➢ Related data type and interface
MI_NIR_IQ_SetSaturation
MI_NIR_IQ_GetSaturation

3.18. MI_NIR_IQ_WeightParam_t

- Description

  Define IQ VIS and NIR Weight parameters.

- Definition

```
typedef struct MI_NIR_IQ_WeightParam_s
{
    MI_U16 u16VisCurveX[WEIGHT_X_NUM];
    MI_U16 u16VisCurveY[WEIGHT_Y_NUM];
    MI_U16 u16NirCurveX[WEIGHT_X_NUM];
    MI_U16 u16NirCurveY[WEIGHT_Y_NUM];
} MI_NIR_IQ_WeightParam_t;
```

- Member

Membername	Description
u16VisCurveX[WEIGHT_X_NUM]	VIS weight curve x coordinate，every value represents the gap between the pre and next value.It is power of 2. It represents the luminance. WEIGHT_X_NUM = 13. Value Range: 0~11
u16VisCurveY[WEIGHT_Y_NUM]	VIS weight curve y coordinate. It represents VIS noise weight. The vaule bigger the weight of vis higher. WEIGHT_Y_NUM = 14. Value Range:0~0xFFFF；
u16NirCurveX[WEIGHT_X_NUM]	NIR weight curve x coordinate，every value represents the gap between the pre and next value.It is power of 2. It represents the luminance. WEIGHT_X_NUM = 13. Value Range:0~11
u16NirCurveY[WEIGHT_Y_NUM]	NIR weight curve y coordinate. It represents NIR noise weight. The vaule bigger the weight of nir higher. WEIGHT_Y_NUM = 14. Value Range:0~0xFFFF；

※ Note

  N/A

- Related data type and interface

  N/A

3.19. MI_NIR_IQ_WeightAutoAttr_t

- Description

  Define IQ VIS and NIR Weight Auto mode struct.

- Definition

```
typedef struct MI_NIR_IQ_WeightAutoAttr_s
{
    MI_NIR_IQ_WeightParam_t  stParaAPI[MI_NIR_AUTO_NUM];
} MI_NIR_IQ_WeightAutoAttr_t;
```

- Member

Membername	Description
stParaAPI[MI_NIR_AUTO_NUM];	VIS/NIR weight auto mode struct. MI_NIR_AUTO_NUM = 16

※ Note

N/A

- Related data type and interface

  N/A

3.20. MI_NIR_IQ_WeightManualAttr_t

- Description

  Define IQ VIS and NIR Weight Manual mode struct.

- Definition

```
typedef struct MI_NIR_IQ_WeightManualAttr_s
{
    MI_NIR_IQ_WeightParam_t stParaAPI;
} MI_NIR_IQ_WeightManualAttr_t;
```

- Member

Membername	Description
stParaAPI;	VIS/NIR manual mode struct

※ Note

N/A

- Related data type and interface

  N/A

3.21. MI_NIR_IQ_WeightAttr_t

- Description
  Define IQ VIS and NIR Weight Attr struct.

- Definition

```
typedef struct MI_NIR_IQ_WeightAttr_s
{
    MI_NIR_IQ_Bool_e                 bEnable;
    MI_NIR_IQ_OpType_e               enOpType;
    MI_NIR_IQ_WeightAutoAttr_t       stAuto;
    MI_NIR_IQ_WeightManualAttr_t     stManual;
} MI_NIR_IQ_WeightAttr_t;
```

- Member

Membername	Description
bEnable	Trigger of VIS/NIR Weight settings. OFF:E_NIR_IQ_FALSE = 0 ON:E_NIR_IQ_TRUE = 1
enOpType	VIS/NIR weight work mode. Auto mode:E_NIR_IQ_OP_TYP_AUTO = 0 Manual mode:E_NIR_IQ_OP_TYP_MANUAL = 1
stAuto	VIS/NIR weight Auto mode struct.
stManual	VIS/NIR weight Manual mode struct.

※ Note
N/A

- Related data type and interface
  MI_NIR_IQ_SetWeight
  MI_NIR_IQ_GetWeight

4. NIR ERROR CODE

API error codes are as the table shown below:

Error code	**Macro definition**	**Description**
0	MI_NIR_OK	Success
-1	MI_NIR_ERR	Fail
0xA0272001	MI_ERR_NIR_MOD_NOT_INIT	Module is not initialized
0xA0272002	MI_ERR_NIR_DEV_ INVALID	Invalid device ID

Error code	**Macro definition**	**Description**
0xA0272003	MI_ERR_NIR_NORESOURCE	Device does not have enough resources
0xA0272004	MI_ERR_NIR_CHN_INVALID	Invalid channel ID
0xA0272005	MI_ERR_NIR_TIMEOUT	NIR Device Running Timeout
0xA0272006	MI_ERR_NIR_NOMEM	Failed to allocate memory
0xA0272007	MI_ERR_NIR_NOTSUPPORT	Not Supported
0xA0272008	MI_ERR_NIR_BUSY	Channel system is busy
0xA0272009	MI_ERR_NIR_NOTREADY	Device is not initialized
0xA027200A	MI_ERR_NIR_NULL_PTR	Null pointer parameter
0xA027200B	MI_ERR_NIR_CHN_NOT_CREATE	Channel not created.